TE WHARE WĀNANGA O TE ŪPOKO O TE IKA A MĀUI
VICTORIA
UNIVERSITY OF WELLINGTON
SCHOOL OF ECONOMICS
AND FINANCE

ECON412 INTERNATIONAL ECONOMICS: TRADE

TRIMESTER 2, 2018

LECTURE WEEK #12: TRADE POLICY

16 OCTOBER 2018

OVERVIEW

- **Readings:** Feenstra Chapter 8.
- General framework on the effects of import tariffs.

1. OVERVIEW OF TRADE POLICY

- Trade policies can affect prices, resource allocation, and income distribution.
- Instruments of trade policy
  - Tariffs.
  - Quotas.
  - Regulations: standards (e.g., health, industrial), labeling.
  - Free trade agreements.
- Trade policies intended to promote domestic growth and development
  - Import substitution: to protect infant industry.
  - Export promotion.
- Trade policies affect welfare.
- GATT/WTO
  - Escape clauses: allow for temporary protection when domestic industries are experiencing unusual competition from imports.
  - Anti-dumping duties: tariffs imposed to offset import prices that are "too low".

2. SOCIAL WELFARE UNDER IMPORT TARIFFS

- Consumer h in a country has a quasi-linear preference, $c_0 + U^h(c^h)$, where c_0^h is the consumption of a numeraire good and c^h is the consumption vector of all other goods. $U^h()$ is increasing and strictly concave.
- Budget constraint for consumer h: $c_0^h + p'c^h \leq I^h$.
- Let $c^h(p)$ be the vector of optimal consumption for consumer h, spending on the numeraire good is then $c_0^h = I^h - p'c^h(p)$.
- Define the social welfare function as the sum of individual indirect utilities:
$$W(p, I) \equiv \sum_{h=1}^{H} I^h - p'c^h(p) + U^h(c^h(p)).$$
By the envelop theorem, the derivative of the social welfare function w.r.t. price is the negative total consumption, i.e., $\partial W / \partial p = \sum_{h=1}^{H} -c^h(p) \equiv -c(p)$.
  - $c(p)$ is a vector aggregate consumption of non-numeraire goods.

- Further simplify the analysis by imposing following assumptions.
  - There is only one single imported good subject to a tariff, holding prices of all other goods fixed.
  - Then p is a scalar.
  - The world price is p^*, the tariff is t, then $p = p^* + t$.
  - The numeraire good is also traded with a world price of being normalized to one.
  - Labor is the only factor of production.
  - Producing one unit of the numeraire good requires one unit of labor, implying that *wage is one*.
- These assumptions allow the economy to mimic a partial-equilibrium setting, where wages are fixed and trade is balanced through the flows of the numeraire good.

- The cost of producing y units of the good in question is $G(y)$. Amount of imports is $m = c(p) - y$.
- The tariff revenue, tm, is redistributed to consumers. Profits from the import-competing industry, $\pi = py - G(y)$, is also distributed to all consumers.
- The social welfare is then written as
$$W[p, L + tm + py - G(y)] \equiv W(t).$$

- The change in social welfare with the tariff is
$$\frac{dW(t)}{dt} = t\frac{dm}{dp}\frac{dp}{dt} - m\frac{dp^*}{dt} + [p - G'(y)]\frac{dy}{dt}.$$

- To derive this result, we have used 1) $\frac{dW}{d(L+tm+py-C(y))} = 1$ (derivative of indirect utility with respect to income is 1 under quasilinear utility); 2) $dW/dp = -c(p)$; 3) $m = c(p) - y$; and 4) $p = p^* + t$.

- The effect of a change in tariff on social welfare is reflected in its effect on domestic price, on foreign price, and on markup.

- The above general form of effect can be more specific, depending on both domestic and foreign market structures.

- We consider three cases.

- **Case 1**, perfect competition, small country.
  - Perfect competition: markup equals 0, that is, $(p - G'(y)) = 0$.
  - *Small country*: world price p^* is fixed as the tariff varies, implying that $dp^*/dt = 0, dp/dt = 1$.
  - Then, $\frac{dW}{dt} = t\frac{dm}{dp}$.
- Note, $\frac{dm}{dp} = c'(p) - \frac{dy}{dp}$ and $c'(p) < 0$. Under perfect competition, $p = G'(y)$, then $G''(y)\frac{dy}{dp} = 1$ or $\frac{dy}{dp} = 1/G''(y)$, which is positive. Therefore $\frac{dW}{dt} < 0$, suggesting that the socially optimal tariff should be 0.
- The welfare lose of increasing tariff from 0 to t is approximated as $W(t) - W(0) \approx \frac{1}{2}\Delta p \Delta m < 0$.

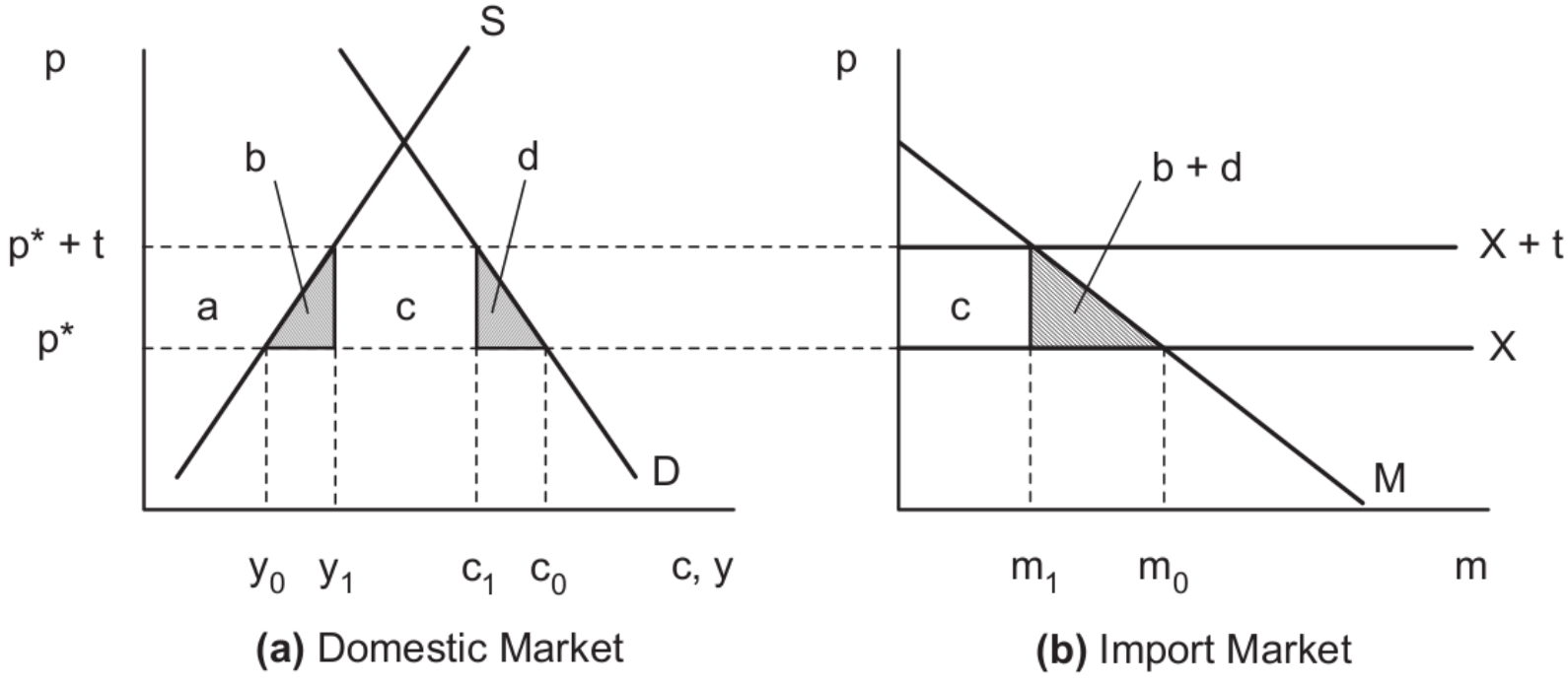

Figure 7.1 Small Country

- **Case 2**, perfect competition, large country.
  - Perfect competition: $(p - G'(y)) = 0$.
  - *Large country*: world price p^* changes with the tariff.
- Then, $\frac{dW}{dt} = t\frac{dm}{dp}\frac{dp}{dt} - m\frac{dp^*}{dt}$.
  - Two effects of tariff: marginal deadweight loss (the first term); the terms of trade effect (the second term).

- Claim: $\frac{dp^*}{dt} < 0$. import price becomes lower as tariff increases. When facing the large home country, exporters have a positive sloped export supply curve X, this together with the downward sloped import curve implies that $p^* + t$ increases by a smaller amount than the increase in t. Therefore, the claim. See Figure next page.

- Then, $\frac{dW}{dt}|_{t=0} = -m\frac{dp^*}{dt} > 0$. Optimal tariff should be positive.

- Setting $\frac{dW}{dt} = 0$ gives the optimal tariff, $\frac{t}{p^*} = \left(\frac{dp^*}{dt}\frac{m}{p^*}\right) / \left(\frac{dm}{dt}\right) = \frac{dp^*}{dm} \cdot \frac{m}{p^*}$. In equilibrium, imports equal foreign country's exports, then $\frac{t}{p^*} = \left(\frac{dx}{dp^*}\frac{p^*}{x}\right)^{-1}$.

- **The optimal tariff (as a percentage of import price), t/p^*, equals the inverse of the elasticity of foreign export supply**.

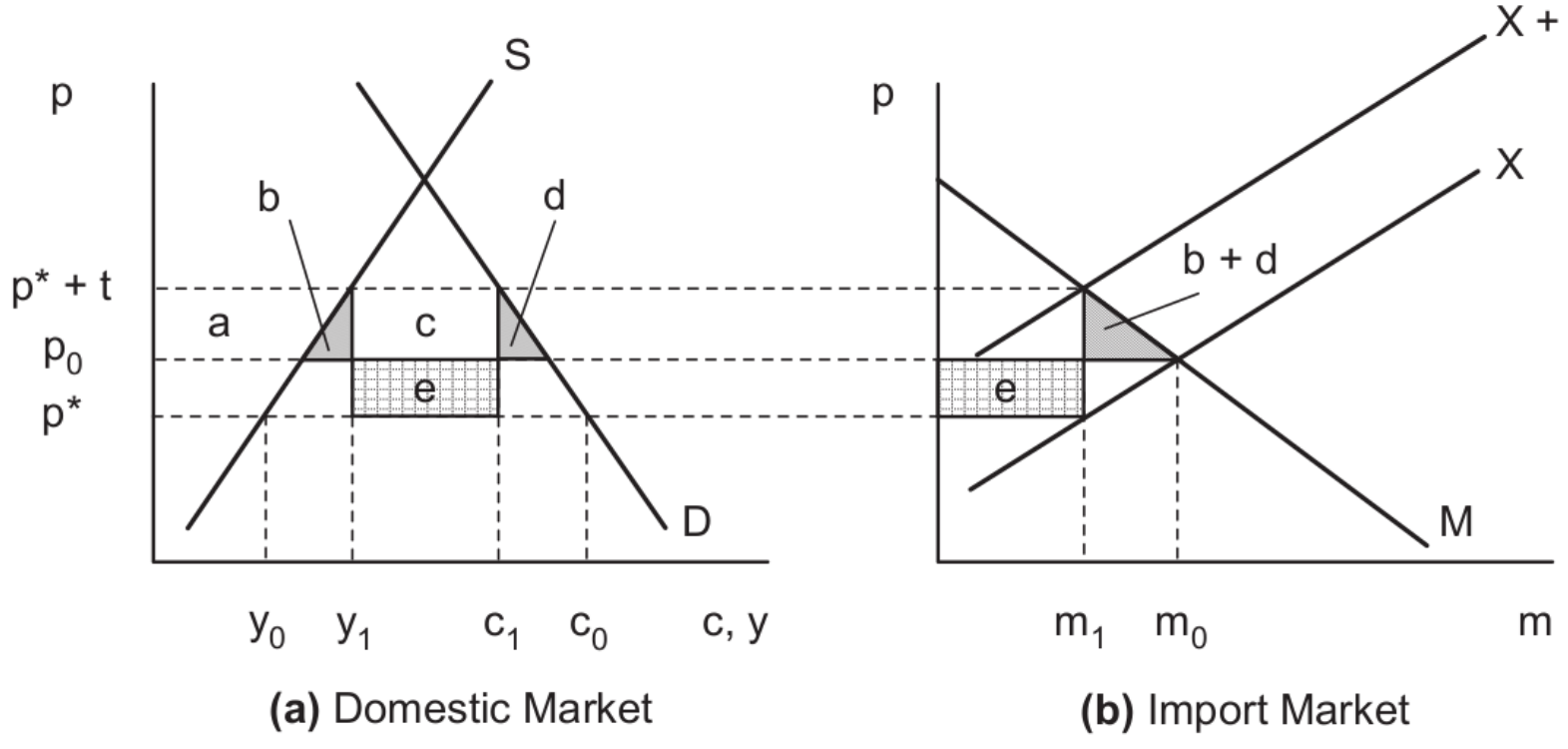

Figure 7.2 Large Country

- Estimation of the elasticity of foreign export supply, Broda etal (2008).

TABLE 3A—INVERSE EXPORT SUPPLY ELASTICITY STATISTICS

Statistic	Observations[a]	Median[b]			Mean		Standard deviation	
Sample	All	Low	Medium	High	All	W/out top decile	All	W/out top decile
Algeria	739	0.4	2.8	91	118	23	333	47
Belarus	703	0.3	1.5	61	85	15	257	36
Bolivia	647	0.3	2.0	91	102	23	283	49
China	1,125	0.4	2.1	80	92	17	267	35
Czech Republic	1,075	0.3	1.4	26	63	7	233	18
Ecuador	753	0.3	1.5	56	76	13	243	30
Latvia	872	0.2	1.1	9	52	3	239	8
Lebanon	782	0.1	0.9	31	56	7	215	18
Lithuania	811	0.3	1.2	24	65	6	235	16
Oman	629	0.3	1.2	25	209	7	3,536	21
Paraguay	511	0.4	3.0	153	132	67	315	169
Russia	1,029	0.5	1.8	33	48	8	198	18
Saudi Arabia	1,036	0.4	1.7	50	71	11	232	25
Taiwan	891	0.1	1.4	131	90	20	241	43
Ukraine	730	0.4	2.1	78	86	16	254	34
Median	782	0.3	1.6	54	85	13	243	30

- Estimation of the relation between exporter size and the foreign export elasticity, Broda etal (2008).

TABLE 6—INVERSE EXPORT SUPPLY ELASTICITIES, GDP, REMOTENESS, AND IMPORT SHARES

Dependent variable	Log inverse export supply		
Log GDP	0.17	0.18	
	(0.04)	(0.03)	
Log remoteness		0.40	
		(0.15)	
Share of world HS4 imports			7.19
			(1.48)
Observations	12,343	12,343	12,343
R^2	0.26	0.26	0.25
R^2 within	0.01	0.02	0.00

Notes: All regressions include four-digit HS fixed effects (1,201 categories). Robust standard errors in parentheses. In the log GDP regressions, standard errors are clustered by country. GDP is for 1996. Remoteness for country i is defined as $1/(\Sigma_j \text{GDP}_j/\text{distance}_{ij})$. The share of world imports is calculated in 2000.

- Estimation of the relation between tariff and the foreign export elasticity, Broda etal (2008).

TABLE 8—TARIFFS AND MARKET POWER ACROSS GOODS (WITHIN COUNTRIES): IV ESTIMATES

Dependent variable	Average tariff at four-digit HS (%)								
Fixed effects	Country			Country and industry			Industry by country		
Estimation method	IV GMM (1)	IV GMM (2)	IV GMM (3)	IV GMM (4)	IV GMM (5)	IV GMM (6)	IV GMM (7)	IV GMM (8)	IV GMM (9)
Inverse exp. elast.	0.040			0.089			0.075		
	(0.027)			(0.055)			(0.028)		
Mid and high inv. exp. elast.		3.96			8.88			9.07	
		(0.76)			(1.18)			(1.08)	
Log(1/export elasticity)			0.75			1.71			1.73
			(0.15)			(0.23)			(0.21)
Observations	12,258	12,258	12,258	12,258	12,258	12,258	12,258	12,258	12,258
No. of parameters	16	16	16	35	35	35	284	282	283
1st stage F	5	1649	1335	2	653	517	3	691	544

Notes: Standard errors in parentheses (heteroskedasticity robust). Industry dummies defined by section according to the Harmonized Standard tariff schedule.

- **Case 3**, Cournot duopoly: one domestic firm competing with one foreign firm in the domestic market. Each firm chooses the optimal amount to sell. A price adjusts to clear the market.

- Let x be the sales of foreign exporter in domestic market, and y be the sales of domestic firm. Total market supply is $z = x + y$. In equilibrium, it equals to demand, that is, $z = c(p)$. The inverse demand is $p = p(z)$ with $p'(z) < 0$.

- Profits of domestic and foreign firms are respectively, $\pi = yp(z) - G(y)$ and $\pi^* = x[p(z) - t] - G^*(x)$. The first-order necessary conditions are respectively
$$\pi_y p(z) + yp'(z) - G'(y) = 0;$$
and
$$\pi_x^* = p(z) + xp'(z) - [G^{*'}(x) + t] = 0.$$

- The second-order conditions are $\pi_{yy} = 2p'(z) + yp''(z) - G''(y) < 0$ and $\pi^*_{xx} = 2p'(z) + xp''(z) - G^{*''}(x) < 0$. The stability condition is $|\pi_{yy}\pi^*_{xx}| > |\pi_{yx}\pi^*_{xy}|$, with which the game between two firms achieves the Nash Equilibrium $(\tilde{x}, \tilde{y})$ no matter where they start.

- The first-order conditions yield best reaction functions $y = r(x)$ and $x = r^*(y, t)$. At the intersection of the two functions the game reaches the Cournot equilibrium, see Figure next page.

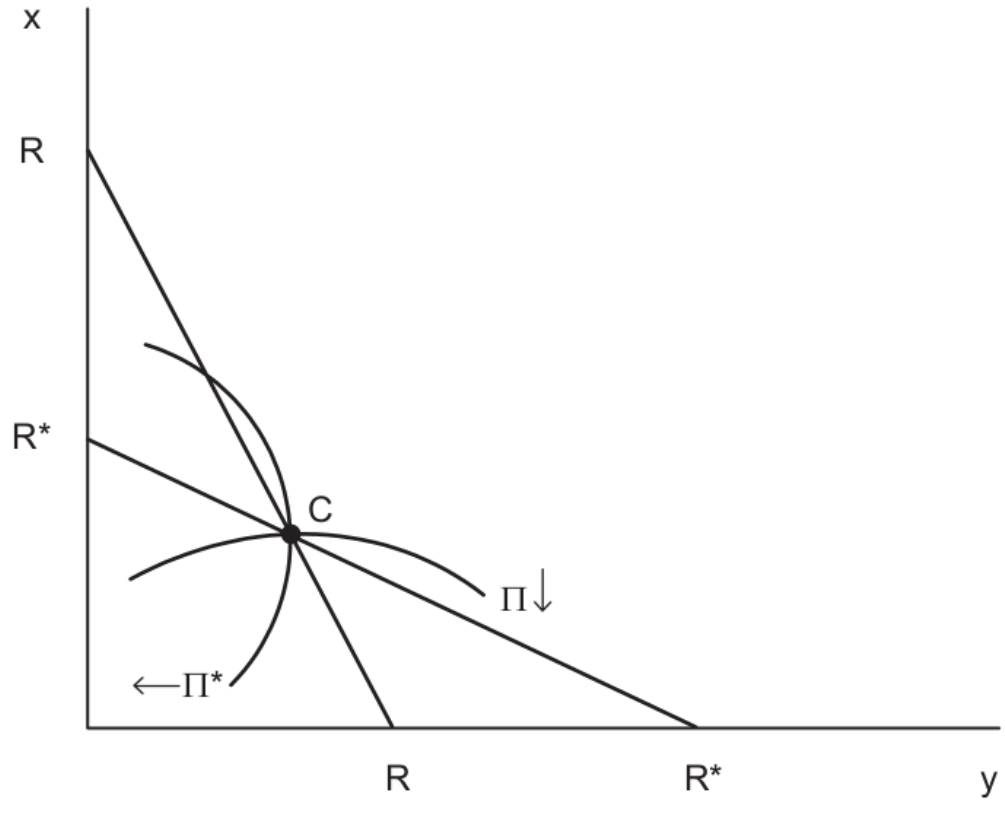

Figure 7.4

- Effect of tariff, from the first-order condition of foreign exporter, we know that $dx/dt = 1/\pi^*_{xx} < 0$. This implies that foreign firm will reduce exports if tariff rises, its reaction function will shift downward.

- As tariff rises, domestic firm produces more (y becomes larger), and foreign firm reduces exports (x becomes lower). How does $z = x + y$ change? One way to show the change is to add the two first-order conditions, we have
$$2p(z) + zp'(z) = G'(y) + G^{*'}(x) + t.$$
Suppose that marginal costs are constant for both firms. differentiating the above equation with respect to t, we have $\frac{dz}{dt} = \frac{1}{3p'(z)+zp''(z)}$ and $\frac{dp}{dt} = \frac{p'(z)}{3p'(z)+zp''(z)}$ (for the latter, since $dp(z)/dt = dp(z)/dz \cdot dz/dt$). Whether these derivatives are positive depends on $(3p'(z) + zp''(z))$.

- $\frac{dz}{dt} < 0$ if $(3p'(z) + zp''(z)) < 0$. Further consider if $\frac{dp}{dt} < 1$. for this to hold, we must have $p'(z) > 3p'(z) + zp''(z)$, where we note that $p'(z) < 0$. This in turn requires $2p'(z) + zp''(z) < 0$, or $p'(z) + zp''(z) < 0$ since $p'(z) < 0$. The last condition says that the total marginal revenue, $p(z) + zp'(z)$, is *downward slopping*, which is satisfied by most demand curves.

- The social welfare in Case 3, by noting that $x = m$,
$$\frac{dW}{dt} = t\frac{dx}{dt} - x\frac{dp^*}{dt} + [p - G'(y)]\frac{dy}{dt}.$$
  - The first term vanishes if t is small.
  - $\frac{dp^*}{dt} < 0$ (terms-of-trade effect) and $\frac{dy}{dt} > 0$ (efficiency effect), from the above analysis.
  - Then $\frac{dW}{dt} > 0$, a positive is welfare improving.
- Note: $\frac{dy}{dt}$ can be negative, depending on the reaction function $y = r(x)$.

- **Other cases**
  - Bertrand competition in duopoly: the domestic firm and the foreign firm compete by setting prices. If the imports x (or m) and the domestic good (y) are perfect substitute, the optimal strategy is to set price to the marginal cost for both firms. In the textbook, Feenstra allows that the imports and domestic good are imperfect substitute. Under some additional assumptions, tariff is welfare improving.
  - Foreign monopoly: a single foreign producer is the only supplier of the non-numeraire good, no competition from domestic. This is an extreme case. Under some conditions, a tariff can improve social welfare.

3. RECIPROCAL DUMPING [SELF-STUDY]

- Oligopoly: N firms in the home country and N^* firm in foreign country. Cournot competition: firms choose the quantity to sell, equilibrium price clears the market.

- Domestic firm i sells y_i and y_i^* respectively in domestic and foreign markets. Foreign firm j sells x_j to other countries and x_j^* to its local market. The inverse demand is $p(z)$, with $z = \sum_{i=1}^{N} y_i + \sum_{j=1}^{N} x_j$. The inverse demand in the foreign country is $p^*(z^*)$ with $z^* = \sum_{i=1}^{N} y_i^* + \sum_{j=1}^{N} x_j^*$.

- There is an iceberg cost T of delivering a product from one country to the other.

- One can think of that a dumping occurs at home country if $p < Tp^*$ (foreign firm sells its product at a price that is lower than its price in its original local market after taking into account transportation costs), and it occurs in foreign country if $p^* < Tp$. According to this criteria, *reciprocal dumping* occurs if the equilibrium prices equal ($p = p^*$).

- Question of interest: whether reciprocal dumping occurs if market size differs in home and foreign countries.

- Let g be the marginal cost and a be the fixed cost of production in each country. The marginal cost of delivering one unit of good from country i to country j is gT. The home firms solve for the profit maximization problem
$$\max_{\{y_i, y_i^*\}} \pi = [p(z) - g]y_i + [p^*(z^*) - gT]y_i^* - a.$$
The first-order conditions are
$$\pi_{y_i} = p(z) + y_i p'(z) - g = 0,$$
and
$$\pi_{y_i^*} = p^*(z^*) + y_i^* p^{*'}(z^*) - gT = 0.$$

- Do the same thing for foreign firms.
- Then impose one symmetry condition: $y_i = y$ in home market and $y_{i^*} = y^*$ in foreign market.
- The four equilibrium conditions, together with the two market-clearing conditions, give the equilibrium number of firms, N and N^*.
- **Theorem**

> *If the elasticities of demand and marginal costs of production are equal across countries, and firms in both countries sell in both markets, then reciprocal dumping occurs:* $\frac{p}{T} < p^*$ *and* $\frac{p^*}{T} < p$.

4. SUMMARY

- Whether import tariff improves social welfare or not depends on the market structure, as well as the elasticities of demand and the marginal costs.
- Empirical evidence confirms the theory.